

- Use 3×3 Gaussian derivative kernels with standard deviation $\sigma = 0.5$, ensuring the absolute sum of kernel values equals 1.
- The $w(x, y)$ is a 3×3 Gaussian window function with standard deviation $\sigma = 0.6$, normalized so the sum equals 1.
- Use border replication during gradient computation and zero padding during smoothing.
- Report each of these kernels, and smoothed I_x^2 , I_y^2 and $I_x I_y$.

2. Cornerness Response Calculation:

Compute a cornerness measure R for each pixel using the following formulations.

$$R = \det(M) - \kappa * \text{trace}(M)^2$$

- Use $\kappa = 0.04$.
- Display the $\det(M)$, $\text{trace}(M)^2$, and scaled response map with scale range 0 - 255.

3. Thresholding:

Apply following threshold to zero method to the cornerness response R to isolate pixels with high corner potential with $T = \text{mean}(R) + 0.7\text{std}(R)$.

$$R(x, y) = \begin{cases} R(x, y), & \text{if } R(x, y) > T \\ 0, & \text{otherwise} \end{cases}$$

- Display the resulting image.

4. Non-Maxima Suppression (NMS):

Perform 3×3 non-maximum suppression on the thresholded image to retain only the strongest local corner responses. Report the result.

Deliverables

1. **Technical Report:** A comprehensive (as briefly detailed as possible) **handwritten** report documenting the following:

- Introduction and methodological overview.
- Presentation and analysis of all visual results (intermediate and final outputs) with appropriate captions and references.
- A short conclusion summarizing observations, findings, and any encountered challenges.

Note: The report must be original and individually prepared.

Submission Instructions

Submission Deadline: **October 14, 2025, until 5:00 PM** (1 week).

Late Penalty: There will be 05% deduction of marks for each day of delayed submission.

Post-Submission: A formatted Excel sheet will be provided for digital validation of results. Therefore, retain a scanned version of your report.

Additional Notes

- Round down all intermediate values to the nearest integer.
- Do not normalize any data unless explicitly instructed to scale.